

Simulation of a Hypersonic Busemann Inlet at Off Design Mach Number and Angle of Attack

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In this study, three-dimensional simulations of a hypersonic streamtraced Busemann inlet at sub design Mach number, and various angles of attack are performed using a positivity-preserving and entropy bounded discontinuous Galerkin formulation. The inlet is designed for operation at Mach 5.5 and all simulations are run at Mach 3.7 at angles of attack of 0° and $\pm 5^\circ$. The simulations are validated against available experimental pressure sensitive paint measurements. The shock structure and pressure distributions are examined at each angle of attack to identify the change flow structures. Additionally, downstream planes are extracted to determine the mass capture and mass flow lost through the bleed slot under each operating condition. Further analysis will be performed to extract the turbulence characteristics and fluctuations occurring at the exit of the inlet.

I. Introduction

IMPROVED understanding of the flow-field dynamics within high-speed airbreathing propulsion vehicles is necessary for further developing hypersonic flight vehicles. These vehicles require stable and efficient operation to occur over a range of flight Mach numbers and altitudes [1]. Although the thermodynamic cycle of a scramjet engine is relatively simple and well understood, designing an engine to operate across a large range of flight conditions and vehicle attitudes remains a significant challenge. The inlet plays a critical role in vehicle operation, capturing and compressing the incoming air [2]. Common inlet designs, however, are typically optimized for specific flow conditions and suffer performance losses under off-design conditions [3].

The most common inlet type explored is the ramp based inlets. These inlets are typically two dimensional and rely on multiple ramps to compress the flow [4–7]. Additionally, several three-dimensional inlet designs using streamline tracing methods (e.g. Busemann Inlets) have become more intensively researched [3, 8, 9]. These inlets are advantageous as they rely on isentropic compression, which allows them to have superior inlet efficiency due to the improved total pressure recovery ratio [10–12]. However, challenges with these inlet configurations arise as they are designed for a single Mach number and with zero angle of attack. This results in significant unknown behavior for airbreathing propulsion vehicles as it is necessary to maintain consistent and stable operation across different flight and attitude conditions.

Many research efforts have looked into the on design performance of these streamline traced inlets [12–16]. However, fewer studies have investigated off-design performance, particularly the effects of varying Mach number and angle of attack. An experimental study by Di Cristina et. al. used planar laser Rayleigh scattering to examine the internal flow characteristics of a stream traced inlet at on and off-design Mach number conditions [17]. The results indicated that at sub-design Mach numbers the general shock and flow features remained similar and that changing the Reynolds number resulted in thickened boundary layer behavior. Additionally, an experimental study performed by Schram et. al. [9] analyzed a streamlined traced inlet at off design conditions and changes in angle of attack. They showed that the mean

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pressure distribution increased under positive angle of attack orientation and decreased under negative angle of attack organization.

Several numerical studies have been performed examining inlet performance as well. Mcready et. al. simulated a Busemann inlet at sub-design Mach number using multiple codes and turbulence modeling approaches[18]. The simulations were able to capture the large region of flow separation on the inlet forebody and that a Menter-type RANS model captured the flow physics adequately. Jung and Im performed RANS simulations of a sugar scoop inlet at sub-design Mach numbers and varying angles of attack to examine the relationship between Mach number and the angle of attack at which the inlet unstarts [19]. Specifically, they found that operation at lower Mach numbers resulted in unstart at progressively lower angles of attack. Bisek performed unsteady Large Eddy Simulations on the HIFiRE-6 vehicle, which included a streamtraced inlet and examined the effects of off-design operation [20, 21]. It was shown that the off design operation changed the captured mass flow and the pressure distribution along the inlet.

The current work examines three-dimensional simulations of the North Carolina State University hypersonic streamtraced busemann inlet from Schram et. al. [9]. Implicit Large Eddy Simulations (ILES) are performed using an extended version of the JENRE[®] Multiphysics Framework with the discontinuous Galerkin scheme presented by Ching et. al. [22]. The inlet is run at a sub design Mach number of 3.7 at 0° and ±5° angle of attack. The simulations are validated against pressure sensitive paint measurements and schlieren imaging. Furthermore, the mass capture characteristics, pressure recovery, and flow uniformity are examined across the different angle of attacks. Additionally, the turbulent flow characteristics generated throughout the inlet and bleed section are examined.

II. Numerical Methodology

In this study, a fully conservative formulation of the multi-component, Navier Stokes equations [23] are solved, which are defined as

$$\frac{\partial (\rho e_t)}{\partial t} + \nabla \cdot ((\rho e_t + p) \mathbf{v}) = \nabla \cdot \left(\lambda \nabla T - \sum_{i=1}^{N_s} W_i C_i h_i \mathbf{V}_i + \boldsymbol{\tau} \cdot \mathbf{v} \right), \quad (1)$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + p \mathbb{I}) = \nabla \cdot \boldsymbol{\tau}, \quad (2)$$

where ρ is density, $\mathbf{v}$ is velocity, p is pressure, $\mathbb{I}$ is an identity matrix, $\boldsymbol{\tau}$ is the deviatoric stress tensor as defined in Eq. (2.17) in [23], e_t is total energy, λ is conductivity, T is temperature, N_s is the number of species, W_i is molecular weight, C_i is concentration, h_i is enthalpy, $\mathbf{V}_i$ is diffusion velocity, ω_i is the production source term, and subscript i indicates species i . The production source term is calculated from the progress reaction rates for any number of reactions and reaction types. Pressure is determined using the equation of state,

$$p = R^o T \sum_{i=1}^{N_s} C_i, \quad (3)$$

where R^o is the universal gas constant. The species diffusion velocity is defined by

$$\mathbf{V}_i = \frac{D_i}{C_i} \nabla C_i - \frac{D_i}{\rho} \nabla \rho, \quad (4)$$

where D_i is the mass-averaged diffusion coefficient.

These equations are solved using the high-order discontinuous Galerkin method, in which the element-local solution is approximated using polynomials of arbitrary order, denoted p . We employ the HLLC inviscid flux function and the BR2 viscous flux function. Hartmann-type artificial viscosity is used to stabilize shocks, and a linear-scaling limiter is used to guarantee that species concentrations are nonnegative, pressure and temperature are positive, and entropy is bounded. More information on the discretization and stabilization strategies can be found in [24].

A. Computational Geometry and Boundary Conditions

The Busemann inlet utilized in these simulations is the same as described in the experimental work of Schram et. al.[9]. The inlet was designed for Mach 5.5 operation with a frontal diameter of 44 mm with a contraction ratio of 3:1. The leading edge has a uniform 2 mm radius. The total length of the inlet is 277 mm and a 51 mm constant area isolator section is added onto the end of the inlet. The geometry is shown in Figure 1. The inlet is placed within a rectangular

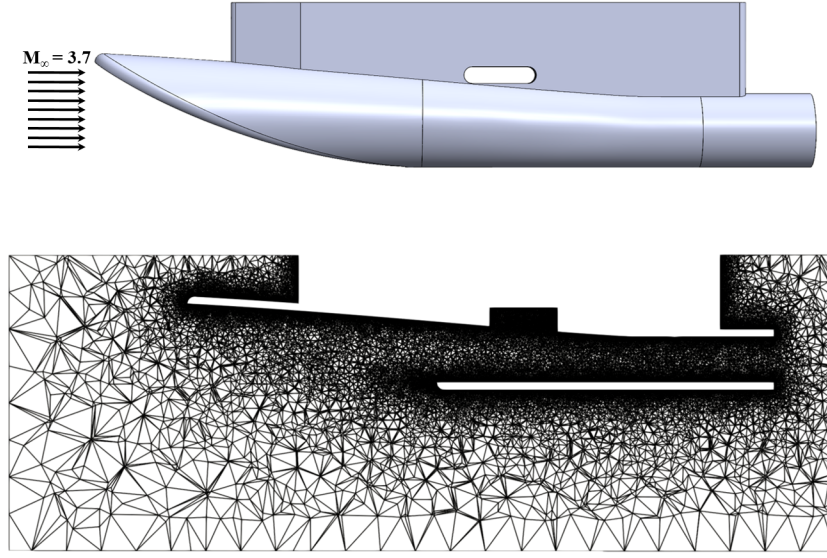


Fig. 1 Geometry and mesh for the 0° angle of attack

test section, which has a length of 650 mm and a 150mm x 150mm cross sectional area. The inlet is able to be run at three separate angle of attacks, which are 0° and $\pm 5^\circ$. The inlet was run at a sub-design Mach number of $Ma = 3.7$ with a freestream pressure and temperature of $P_\infty = 10.9$ kPa and $T_\infty = 79.7$ K, respectively.

The computational domain consisted of the entire wind tunnel test section. In order to reduce cell count, cell clustering is primarily focused on the inlet internal surfaces, leading edge, and bleed slot. The centerplane cross section of the mesh is depicted in Figure 1. The resulting mesh size was approximately 9 million tetrahedral elements and similar sizing was used for all angle of attack conditions. All simulations are presented using third order accurate DG(p=2) elements as this was shown to provide reasonable results on coarse grids in previous supersonic flow simulations [25].

III. Results

The flowfield of the inlet is first examined for each angle of attack. Time averaged contours of the Mach number and pressure are presented in Figure 2. The primary difference in the flowfield is that as angle of attack increases from -5° to $+5^\circ$ the pressure increases from approximately $4 P/P_\infty$ to $15 P/P_\infty$. This is due to the change in flow direction, which results in the air contacting the top wall at the highest angle of attack. This is also seen in the Mach number contours as the core Mach number remain consistent in the -5° angle of attack case. As the angle of attack is increased, the Mach number decreases by nearly half of the freestream Mach number due to the significant pressure rise.

To validate the simulations comparisons are made between the wall pressure and experimental wall pressure measurements made by Schram et. al.[9]. Contours of the top wall pressure distribution for each angle of attack are presented in Figure 3. Similar to the centerplane contours it is seen that wall pressure increases as the angle of attack increases. Additionally, these contours show how the downstream shock wall interactions result in local pressure rises along the wall. The centerline pressure profiles are extracted and compared to the experimental results in Figure 4. At the -5° and -0° angle of attack conditions there is reasonable agreement between the simulations and experimental pressure sensitive paint measurements. At the $+5^\circ$ condition the simulation is over predicting the wall pressure measurements. This behavior is similar to RANS simulation results presented by San Martin et. al. [26].

Additionally, to evaluate the performance of the inlet YZ plane contours are extracted to determine the flow uniformity and characteristics. Instantaneous contours of Mach number and streamwise vorticity (ω_x) are presented in Figure 5. The YZ plane contours are extracted downstream of the bleed slot to quantify the mass flow lost. The captured and mass flow rate lost for each angle of attack is shown in Table 1. The mass flow rate is calculated by calculating the mass flow rate at a plane downstream of the inlet cowl. The mass flow rate lost is determined by comparing the mass flow rate to the downstream YZ planes shown in Figure 5. It is shown that as the angle of attack is increased the

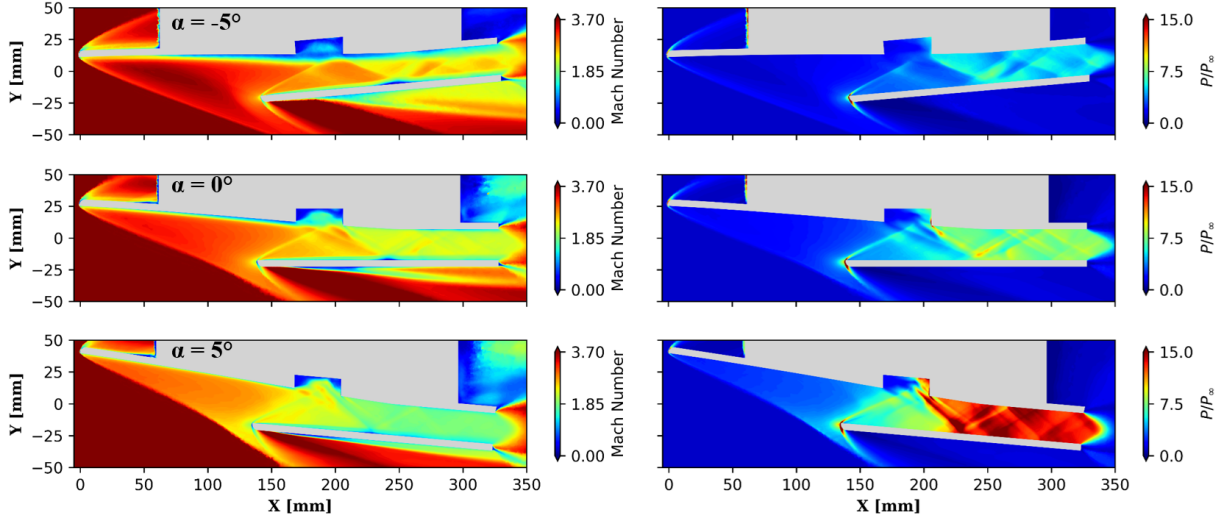


Fig. 2 Time averaged centerplane contours of Mach number and pressure at $\alpha = \pm 5$ and 0

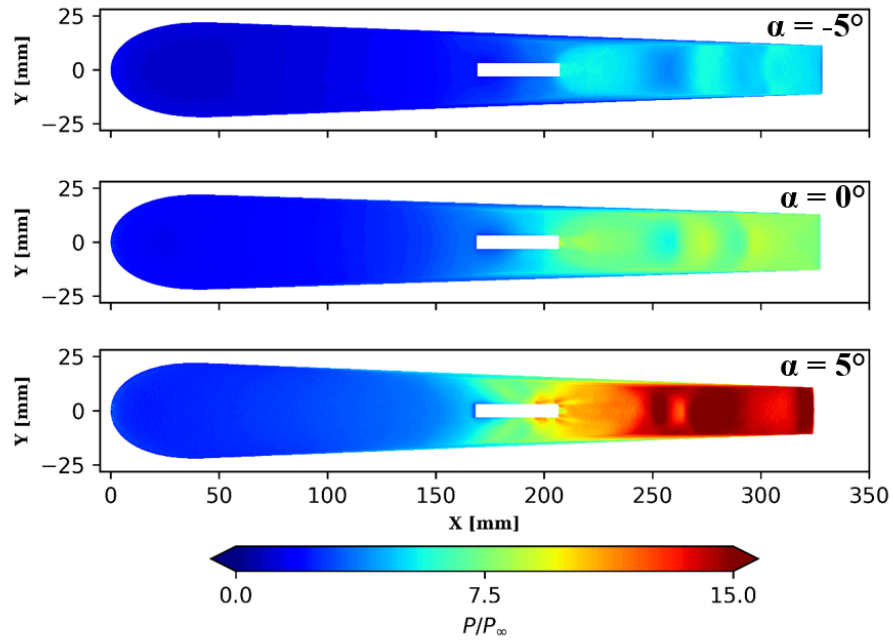


Fig. 3 Time averaged top wall pressure distributions at $\alpha = \pm 5$ and 0

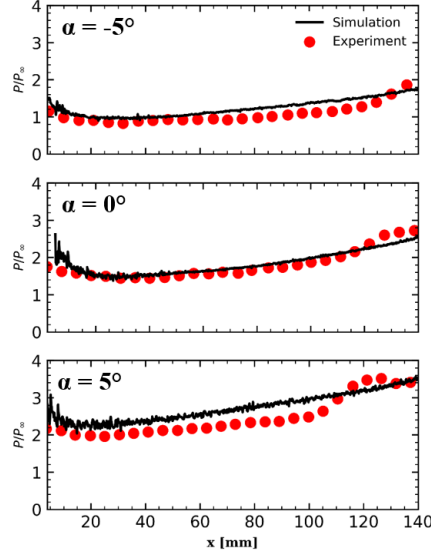


Fig. 4 Time averaged top wall centerline pressure profiles at $\alpha = \pm 5$ and 0

mass capture rises. However, there is minimal difference between the mass flow rate loss of the 0° and $+5^\circ$ angle of attack cases. This is likely a result of the shear layer behavior being similar for these cases as shown in Figure 2.

The instantaneous contours in Figure 5 show the different flow field characteristics within the inlet cross section. It is seen in both the Mach number and streamwise vorticity contours that at the $+5^\circ$ angle of attack condition results in the most turbulent structures within the cross-section. In contrast, it is seen that in the other two cases the high vorticity regions and rollup structures in the Mach number contours are primarily only present near the top wall and diminish significantly near the side and bottom wall region. An additional key difference is that when the inlet is pitched down, there is no disruption of the incoming boundary layer. In both the 0° and $+5^\circ$ case there is a noticeable breakup of the boundary layer structure in the region where the bleed slot is.

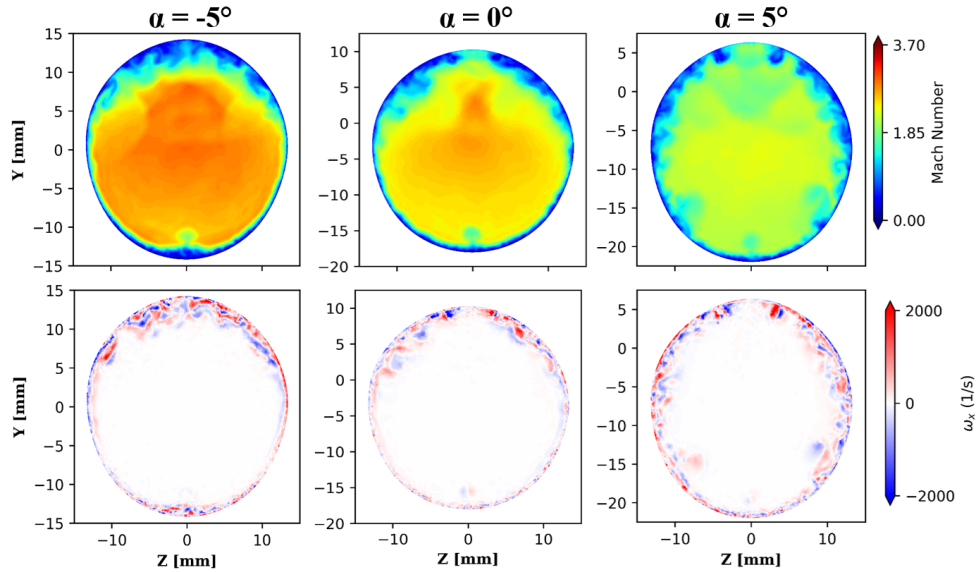


Fig. 5 Instantaneous contours of Mach Number and streamwise vorticity (ω_x) for $\alpha = \pm 5$ and 0

α	-5°	0°	+5°
$\dot{m}$ (kg/s)	0.340	0.455	0.588
$\dot{m}_{loss}$ (%)	3.46	4.61	4.75

Table 1 Mass flow rate captured and mass flow rate lost due at each operating condition

IV. Conclusions

The results of this work will further highlight the influence of angle of attack on operating characteristics of hypersonic streamtraced Busemann inlets. Further analysis will be performed identifying the pressure recovery, flow uniformity, and turbulence characteristics. Additionally, unstart characteristics will be explored through the use of similar back pressure mechanisms as implemented in the experiments.

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